# Vuthtyra (Teera) Yong

Ithaca NY 14850 | vuthtyrayong@gmail.com | https://ty3340.github.io | https://www.linkedin.com/in/vuyong/

## EDUCATION

**Syracuse University, School of Information Studies** Syracuse, NY
M.S. in Applied Data Science, GPA: 4.0/4.0 December 2025
**Relevant Coursework:** Applied Machine Learning, Natural Language Processing, Deep Learning, Advanced Big Data Management, Big Data Analytics, Data Warehousing, Cloud Management, Business Analytics

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, C++
**Machine Learning:** PyTorch, TensorFlow, Hugging Face, LangChain, RAG, supervised/unsupervised learning, time series forecasting
**Data & Big Data:** Apache Spark, Kafka, MongoDB, Redis, Cassandra, Hadoop, Neo4j, Elasticsearch, dbt
**Cloud & Dev Tools:** Azure, AWS, Snowflake, Docker, Kubernetes, Git, Linux
**Visualization & Web:** Tableau, Power BI, Excel, Flask, FastAPI, Streamlit, React

## INTERNSHIP EXPERIENCE

**DENSO (Cambodia) Co., Ltd.** Phnom Penh, Cambodia
Machine Learning Engineer Intern May 2025 – August 2025

- Fine-tuning pretrained YOLO and Faster R-CNN models on manufacturing image datasets for automated defect detection and quality inspection use cases.
- Benchmarked model performance using precision, recall, mAP, and inference speed to identify the most practical solution for real-time inspection workflows.
- Prototyped a containerized inference service using Docker and FastAPI to simulate real-time automated inspection.

## PROJECT

**ScholarAI | React, FastAPI, Azure Blob Storage, Multi-Agent Orchestration**

- Built a full-stack web application that enables users to search arXiv, summarize research papers, generate literature reviews, format citations, and manage saved paper collections.
- Designed a modular agent architecture with remote summarization and shared tool integration to improve maintainability and extensibility.
- Implemented persistent save/load workflows with Azure Blob Storage and added graceful handling of external API failures to improve reliability.

**Multimodal Product Search & Retrieval | Python, SigLIP, PyTorch, FAISS**

- Fine-tuned a SigLIP dual-encoder on a curated subset of the Amazon Berkeley Objects dataset (11K+ products) to align image and text embeddings for semantic search.
- Built an end-to-end retrieval pipeline with FAISS indexes for text-to-image and image-to-text search, achieving **Recall@3 = 0.95** and **MRR = 0.86** on the test set.
- Analyzed learned embeddings with t-SNE visualizations and qualitative search examples to validate stronger cross-modal alignment after fine-tuning.

## CERTIFICATIONS & BADGES

IBM AI Engineering Professional Certificate April 2026
Microsoft Certified: Azure AI Engineer Associate (AI-102) March 2026
AWS Certified Cloud Practitioner (CLF-C02) February 2025
Microsoft Certified: Azure Fundamentals (AZ-900) February 2025
Snowflake Hands-on Essentials: Data Warehouse January 2025